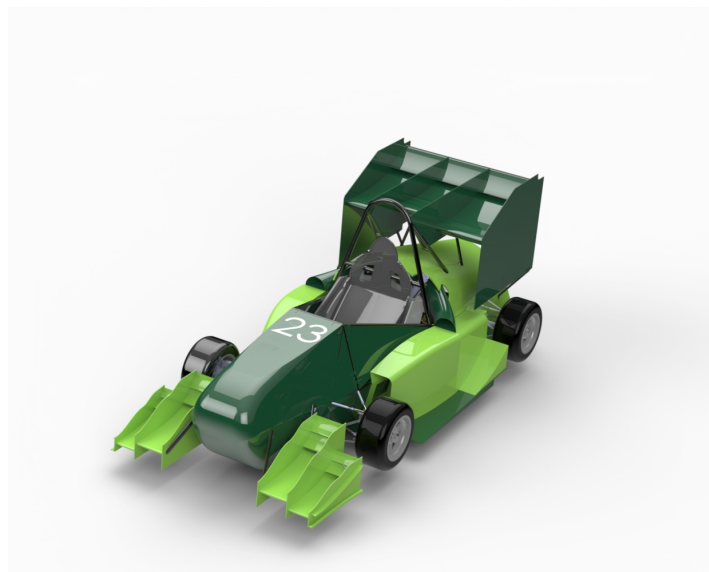

Hybrid Powertrain Global Design, Layout, Strategy and Optimisation

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Abstract

The primary focus of this technical report was to present a comprehensive overview of the design process for a hybrid powertrain intended for the HEV23 Formula Student cars. The final design entails the integration of a KTM 690 motorcycle engine with a 60 kW peak Emrax 188 motor. The gear ratio has been carefully adjusted to achieve a maximum speed of 141 km/h. Various possibilities for enhancing the hybrid system, such as engine charging and regenerative braking, were explored. By employing a Lap simulator, the projected outcome of the final powertrain design indicates a first-place ranking prediction, with a total score of 522.7.

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Executive Summary

Introduction

HEV23 is the proposed design for Imperial Racing Green's Formula Student race car in 2023. The objective is to create a winning car within a budget of 30,000 GBP for the Formula Student 2023 competition. It complies with FSUK regulations and features a hybrid electric and combustion powertrain. With a weight of 276kg, it utilizes a double wishbone pull rods suspension and a steel spaceframe chassis. The car incorporates an aerodynamic package consisting of front and rear wings, a streamlined body, and a diffuser with an undertray. Design decisions were made to meet regulations and were evaluated and validated using the LapSim performance modelling tool.

FSUK regulations have influenced the design process, requiring an understanding of the regulation within each sub-group. Safety, fairness, and compliance with rules are paramount, as failure to adhere can lead to disqualification from dynamic events.

Performance Estimation

The performance team began with an exploration of various lap simulation software before picking the Imperial Lap sim due to its developed physics models and applicability to the Formula Student UK competition. The lap simulator was developed in numerous ways, such as by being expanded to be able to model parallel and through-the-road hybrid configurations along with the capability of regeneration. The lap simulator was also developed to include all events in both simulation and scoring, parallelised to reduce runtime, and via the inclusion of various outputs required by different sub-teams, such as heat power and fuel and energy consumption outputs. All the developments were verified and validated by the performance team. Working alongside other sub-teams – particularly aerodynamics, suspension, and powertrain sub-teams, aiding design development via parameter tuning and optimisation, various sensitivity analysis was conducted across the car to focus the development of the vehicle.

Global Vehicle Layout

The vehicle layout and packaging were determined to maximize points and comply with regulations. The wheelbase was initially set at the minimum regulatory requirement of 1525mm, with a front and rear track of 1200mm based on existing Formula Student cars. The powertrain, chassis, suspension, and aerodynamic package were identified as key components, occupying most of the vehicle's space and mass. The final wheelbase was increased to 1680mm to accommodate the Rear-wheel-drive powertrain system. The front and rear track widths remained at 1200mm for efficient component packaging and to avoid unnecessary weight. The driverless mass of the car was calculated as 276 kg, with the centre of gravity located at within the desired envelope to ensure acceptable handling and performance. The resulting static tire loading was 51.2:48.7 front and rear, respectively.

Chassis Design

The 42 kg chassis was chosen to be made out of a steel frame due to the constraints imposed by the budget, eliminating the possibility of using a carbon monocoque structure. Steel was chosen instead of aluminium for its favourable weldability properties, increasing the flexibility when it comes to adding or modifying elements. The IRG team's standard foam impact attenuator was chosen for its major cost and weight advantages (aside from the fact that it's already available). Extensive finite element analysis simulations have been run in order to ensure compliance and safety of the structure including: frontal impact test, roll test, side impact test and static loading of the chassis (including torsion) under the most limiting track conditions.

Suspension Design

The chosen suspension arm configuration for HEV23 is a double wishbone with a pull rod design. This configuration offers several advantages such as ease of maintenance, excellent handling, stability, and

high tunability. The pickup points were iteratively designed based on optimized target parameters from simulations and dynamic analysis. The dampers were selected using spring rates obtained from lap simulations, allowing for the ideal stiffness to be determined by solving a spring-damper system.

Powertrain Design

The design of this year's hybrid powertrain system aims to make better use of electric power to improve efficiency. The whole group has been worked closely with the performance modelling team to evaluate the performance of past years' hybrid designs.

It was ultimately decided to switch to a new RWD parallel configuration and completely redesign the whole powertrain system including engine, motor, and transmission. To enhance the efficiency of the system, the level of hybridization was increased by integrating a lower-power engine with a more powerful rear-mounted electric motor. This new combination incorporates a compact 52 kW KTM 690cc motorcycle engine and a 60 kW Emrax 188 motor. Both components are connected to a modified input flange on the rear axle differential via two respective chain drive systems.

Furthermore, various hybrid strategies have been investigated. Taking into consideration drivability and feasibility, the team ultimately adopted the 'together strategy'. This strategy utilizes distinct power mappings for the electric motor during different events to prioritize either efficiency or performance. For the endurance event, a 34 kW output mode optimized for efficiency is employed. Conversely, during all other events, the motor operates at full power to achieve the highest scores. Regenerative braking has also been implemented, aiming not only to reduce battery mass but also to increase overall system efficiency. Notably, a significant mass reduction of 40 kg compared to last year's design has been achieved while maintaining a comparable level of power output, ensuring that the hybrid powertrain system stands out among other Formula Student teams in terms of high efficiency and power density.

Battery Selection

A power battery is an essential component in a hybrid race car, it is responsible for storing and supplying the electrical energy required to power the vehicle's electric motor. For this year's design, the chosen approach follows an iterative method, collaborating between the performance and powertrain team. The finalized design incorporates a total of 132 LiPol LPHD8659156 pouch battery cells arranged in a 132-series-1-parallel configuration, resulting in a maximum voltage of 488 Volts. This voltage is used to power the EMRAX 188 490 Volts motor, which has a maximum power output of 60 kW. The battery pack is designed to provide 3.91 kWh of energy, ensuring that the car can complete the 22 km endurance event with a 15% redundancy. The battery pack, consisting of pure cells, has a weight of 15.03 kg and is estimated to cost £900.

Cooling System

This year there was a large focus on designing a more efficient cooling system. By implementing active liquid cooling, the power and torque output of the motor increased by 37% and 40% respectively. Motor controller and electric motors were cooled using the same coolant circuit, with a single radiator in one sidepod. The estimated average total heat output was 2.52kW, with a maximum of 3.22kW. Calculations were based on a 240mm x 120mm radiator, and the optimal mass flow rate of air was determined to be 1.09 kg/s at a car velocity of 20.7 m/s.

The ICE coolant system was designed to prevent the coolant from boiling during the endurance race. The necessary mass flow rate of air was determined, requiring a capture area of 0.047 m², divided between two inlets located above each sidepod. The captured air passes through two radiators to transfer heat from the coolant to the air. The effectiveness of different radiator options were evaluated using the NTU- ϵ method. The final optimized coolant system ensures that the coolant temperature remains below 98.4°C, preventing overheating.

To comply with regulations, the average cell temperature in the tractive system (TS) of the vehicle must not exceed 60°C, with an optimal temperature of 35°C. Channels between the TS modules allow air to be directed through them for cooling. An inlet in the left-hand sidepod guides air through

the TSAC (TS Accumulator Container), which contains the TS cells. Guide vanes are implemented to optimize airflow. An inlet area of 0.020 m² ensures the cell temperature remains below 35.2°C, averaging 33.4°C. The brake discs were also analysed for heat exposure but were found not to require a bespoke cooling system as they would not reach the maximum operating temperature of 600°C.

Aerodynamics

The front wing of HEV23 was designed to optimize the downforce to drag ratio (L/D) and exceed the performance of previous designs. Targets were set to achieve 130.2N of downforce and 23N of drag, resulting in an L/D ratio of 5.66. The front wing consists of an inner and outer triple-element section, with additional features such as reverse 'S' shaped endplates, camber, footplates, and canards to improve aerodynamic performance and minimize drag. Star-CCM+ 2D and 3D Simulations were conducted to fine-tune the design. The final performance evaluation showed a downforce and drag of 112 N and 20 N, respectively. This translated to a decrease of 5.8% in downforce coupled with a 13% decrease in drag compared to the HEV4. The front wing is mounted to the front bulkhead using clamped steel rods allowing for easy removal and replacement. These rods are bonded with epoxy adhesive to the front wing. The wing is made of a quasi-isotropic layup of 2mm thick CFRP surrounding a polyurethane foam weighing 7kg in total.

The rear wing of HEV23 was designed to achieve high downforce efficiently with 164.3 N of downforce and 40 N of drag as targets, resulting in a lift to drag ratio of 4.11. The double element configuration places the main element higher relative to the car, reducing the impact of the bluff body wake from the driver and roll hoop. Sculpted endplates positioned inside the rear tires assist the diffuser and block a portion of the rear wheel wake. Discontinuous profiles and vertical plates help control flow and minimize inefficient downforce production. The rear wing generates a total downforce of 208 N with a drag of 42 N, resulting in a lift to drag ratio of 4.95.

To reduce weight, a sandwich panel composite structure is used, consisting of a wet layup of 8 layers of woven carbon fibre cloth, giving a thickness of 2 mm over a polyurethane foam core. The rear wing is mounted using rods bolted onto the endplates and middle plates, reducing torsion on the chassis. The total mass is 9.96 kg.

The CFRP nose cone's main purpose is to provide a mounting point and casing for the front section of the car and prepares a good flow structure towards the rear of the car. It has to integrate with many components such as the chassis, front wing, undertray, and sidepods. The nose cone design focuses on minimal lift and drag is established for the nose cone, with more emphasis placed on incorporating the nose cone with other components and understanding the flow structures around the nose cone in order to achieve aerodynamic efficiency. The chosen design features a high nose configuration that houses the chassis and directs air into the undertray. It generates 1.86 N of downforce, 5.32 N of drag, and provides a mass flow rate of 0.287 kg/s to the undertray. A gap between the front wings and nose cone allows airflow to the sidepods.

The side bodywork includes sidepods and flip-ups, which must meet cooling and aerodynamic requirements. The sidepod inlet is positioned between the wheels and chassis, just 75 mm behind the rear plane of the wheel. It achieves an average mass flow rate of 1.43 kg/s during the endurance event, surpassing the cooling requirement of 1.09 kg/s and accounting for flow instability and discrepancies. The flip-ups reduced the rear wheel drag by 27%. The side bodywork produced a total of 3.0 N downforce and 2.06 N drag. A thin Aluminium 5005 flip-up is placed beside the sidepods to direct any remaining flow over the wheels, reducing wheel drag.

The undertray is designed to optimize diffuser efficiency while respecting regulation requirements. It starts 103.5mm behind the front hoop, maintaining a 30mm ground clearance and extending to the rearmost horizontal bottom chassis member. The width of the undertray is adjusted to avoid contact with the turning tires, matching the distance between the outboards of the tires when the sidepod begins. It then narrows to stay clear of the keep-out zone near the rear tire. The straight edges reduce manufacturing complexity and costs. The car's rake is set at 2.5 ° and skirts extending to 30mm above the ground are added.

The diffuser is designed to generate high downforce while minimizing drag. It starts at the end of the undertray and extends to 250mm past the rear tire. A 15° expansion angle is chosen based on

previous design reports and simulations. Straight and curved strakes are added to the bottom of the diffuser. The diffuser and undertray achieve a total 86N lift and 4.4N drag. They are manufactured in 2 CFRP layers totalling 3mm to minimise weight.

Steering System

The steering system of HEV23 was designed with a focus on packaging efficiency, weight reduction, and cost-effectiveness while complying with FSAE regulations. A front-wheel-drive steering system was chosen to simplify the design based on its widespread use in Formula Student competitions. A rack-and-pinion steering rack, steering wheel, u-joints, and quick release hub were carefully selected. In-house manufacturing was utilized for parts like tie rods and the steering column to optimize packaging. Steering geometry was designed in collaboration with the suspension team using the Ackermann model approach.

Braking System

The braking system prioritizes front wheel performance due to longitudinal weight shift forward during hard braking. A balance bar adjusts brake bias considering lock-up forces at each axle. Due to the 10-inch rim diameter, the brake design features 7-inch discs with dual-piston calipers for all wheels, with regenerative braking modified brake pedal to allow regeneration from the rear wheels as per powertrain team requirements.

Electrical system integration

HEV23 features two main electrical systems: a low voltage system running on a 12V battery for data monitoring using sensors, and a high voltage powertrain system powered by a series of 132 3.7V Lithium Polymer cells. To enhance driver performance and maximize points, a telemetry system for real-time data collection and transmission was incorporated, along with a live communications radio system. The high voltage powertrain system includes the motor controller and hybrid powertrain motors. The design of the Traction System Accumulator Container (TSAC) was modified based on component layout changes from the previous model and to meet cooling requirements specified by the Cooling Team.

Driver Ergonomics

The ergonomics of the HEV23 was designed by taking a 5th percentile female as the lower limit and the 95th percentile male as the upper limit. The visibility was studied to ensure the driver's line of sight to the steering is within 10° of the forward range of vision. A one-to-one scale model was used to study the seating angle and visibility and compared with theoretical calculations. Methods for adjusting the driver's environment were explored to ensure the driver's safety and comfort. Safety features were selected to comply with FSAE rules while minimising weight and cost.

Firewall Regulatory Requirements

The firewall in a Formula Student race car acts as a heat-resistant protective barrier, shielding the driver from engine heat and fluids and prevents fluid leakage. Positioned behind the seat, the firewall covers the back of the seat and extends to the lower section of the driver's helmet. It must be securely attached to the chassis to prevent fluid passage. A 25mm air gap is maintained between the seat and the nearest traction system for insulation. The firewall utilizes Nomex Honeycomb sandwiched between thin aluminium sheets, balancing insulation, weight, and safety.

3D modelling team

The 3D modelling CAD team played the two important roles of modelling the large majority of components designed for HEV23, and verification of the integration of components as they were included in subsystems assemblies. The CAD team would identify if any components clashed and

would really back to the subteams. This team also designed many of the fasteners and spacers when the subsystem design team did not consider them

Cost analysis

Initially, the budget of £30,000 was distributed among different sub teams, considering both historical trends and new strategic focuses for the project. To keep the cost to a minimal, components were manufactured in-house where possible, utilising the facilities available in the college. Additionally, an emphasis was placed on designing components with cost efficiency in mind, ensuring that expenses were kept to a minimum. For components with similar parameters, a cost-to-mass ratio analysis was conducted, whereby a rule of thumb dictated that a 1 kg increase in mass would result in a decrease of 0.239 points in cost. This analysis helped evaluate whether the performance gains justified the associated increase in cost. Ultimately, the final estimated cost of the vehicle was determined to be £29465.92, which remained comfortably below the initial budget of £30,000 considering some components would actually be recycled from IRG.

Final Vehicle Specification & FSAE Competition Prediction

The final performance of the car was found to be 522.7 points, corresponding to a first position result, relative to the teams that participated in the 2022 FSUK competition. Overall, the team succeeded in designing a winning race car within the regulations and budget.

Table 1: Summary of the design of HEV23

Parameter	Specification
Wheelbase /mm	1680
Wheeltrack / mm	1200
L/D	2.97
Mass / kg	276
Predicted FSUK position	1st position (522 points)
Cost / GBP	29,637.20

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Nomenclature

ECU Engine Electronic Control Unit

EM Electric Motor

GDP Group Design Project

HEV4 The Hybrid car design by 2022 Group Design Project

ICE Internal Combustion Engine

VCU Vehicle Control Unit

1 Introduction

1.1 Motivation&Outline of Work

The objective of this 2023 Group Design Project is to develop a race car (HEV2023) specifically for the Formula Student UK competition, catering to the potential customer, Imperial Racing Green Team. The competition consists of various events such as Acceleration, Endurance, Sprint, and Skid-pad [1], with detailed information provided in the report from the group's technical coordinator, Oscar [2]. The hybrid powertrain is identified as a critical component for HEV2023, which has been completely re-designed from the past years' project. This report primarily encompasses the comprehensive overview of the 2023 hybrid powertrain design process, along with certain supportive work to the performance simulation team.

To initiate, the hybrid powertrain integrates the Internal Combustion Engine (ICE) with the Electric Motor (EM), and the level of hybridization can be denoted by a Hybridization Factor (HF) [3] calculated from the power ratio in equation 1.1. :

$$HF = \frac{P_{ElectricMotor}}{P_{ElectricMotor} + P_{Engine}} \quad (1.1)$$

1.2 Appraisal of 2022 Design

The 2022 Design incorporated the 86 kW Triumph Daytona 675 engine [4], which is one of the most powerful engines available within the 710cc displacement regulation. In addition, two 18 kW BLDC front wheel hub motors were utilized to create an All-wheel-drive car, resulting in a potential maximum power output of 122 kW. However, there are certain areas that could be improved this year:

1. Insufficient Hybridization: The Hybridization factor in 2022 was measured at 0.295. It is recommended to increase this factor within the optimal range of 0.3-0.5 to enhance efficiency [5].
2. Inadequate utilization of the electric motor during actual lap running: The 2022 HEV4 adopted a separate Hybrid strategy with a cut-off car speed of approximately 17 km/h. Below this speed, the HEV4 functioned solely as an Electric Vehicle (EV) with Front-wheel-drive, but beyond this speed, it operated solely as a Rear-wheel-drive fuel car. Analysis of the motor power plot (shown in Appendix Figure A.1) revealed minimal activity, indicating that the HEV4 essentially performed as a fuel car during real race events, despite carrying the additional mass of the electric propulsion system.
3. Control and drivability concerns: The powertrain system was designed to utilize either the front motors or the rear engine separately. Consequently, the car never took advantage of the 4WD system, and the mode switch resulted in changes to the vehicle's dynamic performance, potentially making it more challenging to control. Moreover, the power output demonstrated non-linear behavior, as shown in figure A.1, with a power drop observed during the switch.
4. Excessive weight: While the Triumph Daytona 675 engine provided considerable power, it also came with a significant weight of 70 kg. Furthermore, the hub motors, along with their reducers, added 18.7 kg to the overall mass. Due to incorrectly sized battery and underestimation of the wirings, the entire propulsion system was claimed to have a total mass of 102.3 kg. However, upon rectifying the design, the total powertrain mass could exceed 120 kg. For comparison, a typical formula student team employing a pure Internal Combustion Engine (ICE) has main powertrain components weighing around 70 kg (Appendix Figure B.2).

1.3 Objective of 2023 Design

The aim of this project is to design a hybrid powertrain system that fulfills the following objectives:

1. Take the benefits of hybrid technology: The primary objective is to fully leverage the advantages offered by hybrid powertrains, including instant high torque of the electric motor, increased total power output and reduced emissions compared to Internal Combustion Engine (ICE) cars.

2. Mitigate the limitations of hybrid systems: It is crucial to minimize the drawbacks associated with hybrid technology, such as the increased number of components and added weight when compared to traditional ICE cars.
3. Emphasize compactness in powertrain design: The design should prioritize minimizing the space required for the powertrain system, ensuring a more efficient utilization of available vehicle space.
4. Develop an effective hybrid strategy for maximizing performance: A suitable hybrid strategy should be devised to optimize performance and achieve high scores in the Formula Student events.

2 Hybrid Powertrain Development

2.1 Hybrid Layout

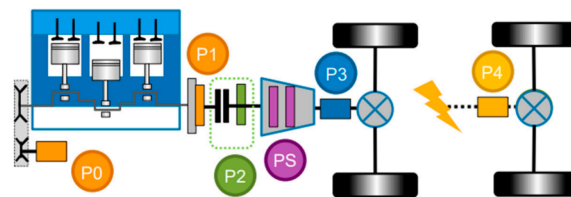


Figure 1: Hybrid system architectures classified by motor position [6]

The Electric Motor (EM) can be integrated with the Internal Combustion Engine (ICE) through electrical/ mechanical coupling methods to configure a 2-Wheel-Drive (2WD) or 4-Wheel-Drive (4WD) Hybrid Vehicle, as outlined below [6]:

1. 2WD Mild or Light Parallel Hybrid by Replacing the 'traditional' Engine starter/generator:
 - P0 Belt-driven Starter/Generator (BSG).
 - P1 Motor connected to the engine crankshaft or flywheel, known as Integrated Starter/-Generator (ISG).

Discussion: The motor speed will always be synchronised with the engine rpm. An example of a similar design used in FS cars can be observed in the Modena Racing Team, where they connected a 30 kW motor to the chain of Valve-Timing (VT) system of a 955cc Ducati motorcycle engine [7]. The motor's torque will be combined with the engine torque on the engine output shaft, and subsequently transmitted to the gearbox. This integration involved a significant level of complexity on engine modification and crankshaft/gearbox enhancement.

2. 2WD Parallel Hybrid with EM placed between the ICE and gearbox:
 - P2 Motor connected to the input shaft of the gearbox and decoupled from the engine by a clutch.

Discussion: Within this configuration, the engine and motor possess the capability to operate collaboratively or autonomously, enabling the motor's torque to be optimally scaled through different gears inside the gearbox. Nevertheless, in the majority of motorcycle powertrains, the gearbox and engine are designed as an integrated unit, leaving no available space for the inclusion of a motor between them. The only way to achieve this is to design a new gearbox with two input shafts connecting both the engine and motor.

3. 2WD Parallel Hybrid with EM placed after the gearbox:
 - P3 Motor installed at the output shaft of the gearbox or connected to the same driving axle powered by the ICE.

Discussion: This arrangement requires no structural alterations to the motorcycle's engine-gearbox unit, providing increased flexibility in choosing electric motor with different dimensions. However, once the motor is connected to the axle beyond the gearbox, the electric motor's speed becomes directly linked to the wheel speed, rather than the engine's. As a result of the electric motor's constant torque characteristics, this setup maintains nearly constant torque augmentation for the ICE-gearbox's torque output at any gear engaged. Furthermore, this configuration enables the ICE and EM to operate jointly or independently by disengaging the clutch inside the motorcycle gearbox. A similar design has been observed in the Team Uttejit Formula Student car [8] with moderate level of complexity.

4. 4WD Parallel Hybrid with EM installed at different driving axle:

- P4 Motor directly drives the wheel on the other drive shaft.

Discussion: The HEV4 design by GDP 2022 has the same configuration. However, due to spatial constraints within the front chassis, the EM must be positioned at the two front wheel hubs, resulting in increased unsprung mass. This added weight may potentially hinder the suspension dynamic performance during laps. Furthermore, the options for hub motors are limited, one of the most popular choices for FS teams is the AMK hub motor, their high price of approximately 25,000 GBP for a pair poses a significant over-budget [9].

5. Other More Complex Compound configurations:

- PS: ICE and EM are connected by planetary gears, named power split (PS) technology, e.g. Toyota Prius.
- Combination of motors: e.g. P0+P4 on BMW i8, P1+P3 on Honda Accord, P1+P4+P4 on Acura NSX.

Discussion: This arrangement offers extensive design flexibility for hybrid mode, enabling independent control of the ICE and EM to operate within their respective optimal working regions, thereby maximizing efficiency [10]. The vehicle can seamlessly switch between series and parallel modes based on specific driving conditions. However, incorporating a planetary gearbox to the transmission system or integrating an additional P0/P1 motor into the engine requires substantial engineering effort, adds mass and complexity to the overall vehicle, and necessitates the development of a sophisticated hybrid system strategy. Due to the time and budget constraints of this GDP project, this layout will not be considered.

As discussed above, considering the factors of performance, efficiency, compactness, and complexity, this year's design philosophy has led to the selection of a RWD Parallel Hybrid layout featuring a P3 motor. As comparison, a detailed analysis of this significant configuration change has been done with the help of the simulation team, in order to assess the benefits of this RWD parallel layout over the past years' 4WD layouts with P4 Motor. For more detail, please refer to Aaditya Aaditya's [11] and Stephen Cao's report [12].

To ensure optimal powertrain integration, the internal combustion engine (ICE) and electric motor (EM) are connected to the rear axle through two distinct chain drive systems. This choice was made to prioritize simplicity and redundancy. For the powertrain integration details, please refer to Shiven's report [13].

2.2 Hybrid System Initial Power Sizing

As per the regulations set by FSUK [14], the hybrid vehicle must follow the requirements specified for both electric vehicles (EV) and internal combustion engine (ICE) cars. These include:

- The use of a 4-stroke internal combustion engine with displacement below 710cc.
- Limiting the power output of the electric motor to 60kW for 4-wheel drive (4WD) configurations and 80kW for 2-wheel drive (2WD) configurations.

- Ensuring driver diversity by considering a weight range from 49kg for 5% female participants to 105kg for 95% male participants.

Table 2: Powertrain Analysis of 2023 FSUK Comparable Teams and past HEV Designs

	2022 FSUK Winner Glasgow UGR [15]	FS Team Uttejti [8]	MoRe Modena Racing Hybrid [7]	2022 HEV4 [16]	2019 HEV2 [17]
Car mass approx. (kg)	210	250	250	262	248
Engine Power (kW)	48	30	46	86	86
Motor Power (kW)	0	30	32.5	18+18	23+23
Max Total Power (kW)	48	60	78.5	122	132
HF	0	0.5	0.41	0.3	0.35
Power/Mass (kW/kg)	0.23	0.24	0.31	0.47	0.53
Average Power/Mass (kW/kg)	0.26			0.50	

As indicated in Table 2, the analysis of previous year’s GDP research and examination of competitive teams’ vehicles in the 2023 FSUK suggest that hybrid vehicles are expected to have increased mass due to the inclusion of an additional electrical propulsion system. However, they also exhibit higher maximum total power output.

In the 2019 and 2022 designs [17][18], a driver weighing 68 kg was selected. However, designing for extreme conditions, the ergonomics team has opted for a 105 kg driver (95% male) [19], which is equivalent to a 37 kg mass addition on the vehicle.

The weight and packing team has informed that the previous GDP design significantly underestimated the total weight, and the aerodynamics team plans to incorporate additional and heavier aerodynamic kits in this year’s design (estimated 10 kg increase with a conservative approximation). This leads to a minimum equivalent mass addition of 47 kg for the HEV2023, resulting in a total equivalent mass of 309 kg for the car. As recommended [5], a hybrid factor (HF) ranging between 0.3 and 0.5 offers the optimal level of hybridization to maximize fuel efficiency and performance.

Table 3: ICE & EM Power Sizing for different Power/Mass and HF

		HF=0.3		HF=0.5	
Power/Mass (kW/kg)	Total Power (kW)	Motor Power (kW)	Engine Power (kW)	Motor Power (kW)	Engine Power (kW)
0.23	71.1	21.3	49.7	35.5	35.5
0.31	95.8	28.7	67.1	47.9	47.9

As illustrated in Table 3 above, for a relatively high power-to-mass ratio of 0.31 (same as Modena Team), the desirable power range for the electric motor lies between 28.7 kW and 47.9 kW, while the engine power ranges from 67.1 kW to 47.9 kW for HF values of 0.3 and 0.5, respectively.

The initial power sizing serves as a starting point for the selection process of the engine and electric motor. For more information on the selection and design of the electric motor, controller, and battery, please refer to Junjie (Jasper) Zhuge’s report [9].

2.3 Engine Selection

As a result of the 4-stroke, 710cc displacement regulation, previous research [4][17][18] has demonstrated the significant benefits of utilizing a motorcycle engine, which offers advantages in terms of both compactness and high power density. Table 4 presents a list of popular engines for FS events, ranked based on their power-to-mass ratios.

In previous GDP designs, the Triumph Daytona 675R engine, known for being one of the most powerful engines compliant with FS regulations, was consistently chosen. This engine boasts the highest power density and an impressive RPM limit of up to 14,000. However, as discussed in the Initial Power Sizing section, the power output of the Triumph engine falls outside our desired range,

resulting in excessive overpower. While it may be an excellent choice for a pure internal combustion engine (ICE) vehicle, it may not be as suitable for a hybrid configuration.

Table 4: Comparison of potential motorcycle engines

Engine Model	Displacement (cc)	Mass (kg)	Peak Power (kW)	Power/Mass (kW/kg)	No. of cylinders	Price Approx.	Transmission
2022 Design Triumph Daytona 675R [20]	675	70	93.6	1.34	3	1800	6MT
KTM 690 Duke [21]	690	40	52	1.30	1	1500	6MT
Honda CBR600RR [22]	599	58	75	1.29	4	1500	6MT
KTM 390 Duke [23]	373	36	32	0.80	1	1200	6MT
Rotax 650 [24]	652	79	45	0.57	2	ltd avail	CVT
Rotax 450 [25]	427	58	28	0.48	1	ltd avail	CVT

Consequently, the KTM 690 Duke's LC4 engine has been selected. This engine offers a moderate power output and an excellent lightweight design. In comparison to other engines, its large single cylinder design delivers substantial torque even at low RPMs. Additionally, its smaller size (compared to the Triumph 675) provides extra space for integrating the P3 hybrid motor alongside the engine at the rear of the chassis, without increasing the overall width of the vehicle too much.

The KTM engine utilizes an electronic control unit (ECU) to achieve precise control over critical engine parameters, including air intake, fuel injection, and ignition. To enable seamless control and communication between the engine and the Vehicle Control Unit (VCU) in the HEV2023, the decision has been made to replace the original KTM ECU by a race-bred Motec M800 ECU. This specific ECU selection will provide extensive engine tuning capabilities, allowing for optimal performance and functionality [26].

3 Engine and Motor Performance Modeling

Following the careful engine and motor selection process, the team has opted for a KTM 690 engine paired with an Emrax 188 HV PMS motor [9]. To accurately model the powertrain's performance and efficiency, as well as provide heat power approximations to the cooling team, extensive research has been conducted to gather reliable performance data for both the engine and motor, which has been given to the performance team for LapSim car model development.

3.1 Engine Modeling

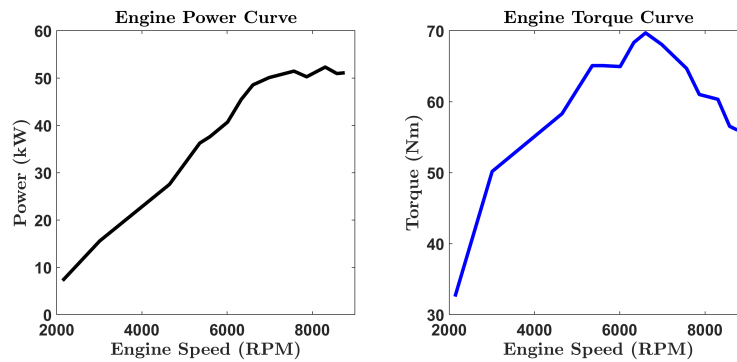


Figure 2: KTM690 Engine Power and Torque Curve

As shown in Figure 2, the engine power and torque versus RPM curve has been obtained from the dynamometer (dyno) curve of the KTM 690 Duke motorcycle (shown in appendix Figure D.2) using

graphical techniques. Since the dyno curve represents the power on wheel, the data already accounts for transmission losses between the gearbox and chains. This data can be directly utilized by the simulation team to analyze the on-wheel output.

The determination of the efficiency mark is directly influenced by CO₂ emissions, which are closely tied to engine fuel consumption. However, accurately quantifying engine fuel consumption is a relatively complex task. In previous GDPs, this was either approximated using the MPG (miles per gallon) data of the original motorcycle or by assuming an engine thermal efficiency and then calculating fuel consumption data in reverse. Unfortunately, the 2022 GDP falsely claimed a fuel consumption of only 1.85 L for the Triumph engine, which significantly underestimated the actual value, leading to incorrect efficiency scores and potentially influencing decision-making.

This year, the data has been extracted from the engine ECU fuel map, which provides the fuel-air mixture mass injected for each stroke at different throttle levels and RPMs. As mentioned in the hybrid layout Section 2.1, a comparison has been made between HEV23 and the 2022 HEV4 using corrected data. This includes a new fuel curve calculated based on the air-fuel mixture readings from both the Triumph and KTM engine ECUs, the calculation of fuel data at each point on the map are briefly explaining below:

The Fuel map readings shown in Appendix (Figure C.1) represents the amount of air-fuel mixture injected to the cylinder times by a scaling factor (which is around 15 for KTM and 20 for Triumph [27]) in mg/stroke.

The desired air-to-fuel (AFR) mass ratio for the KTM 690 engine is 13 [27]. Similarly, the AFR for the Triumph engine also hovers around 13. Generally, a gasoline combustion process achieves complete burning at an AFR of 14.7. Deviations from this value indicate different operating conditions. A lower AFR, below 14.7, suggests a richer mixture with more fuel injected, leading to enhanced performance. Conversely, a higher AFR indicates a leaner mixture, allowing more air to participate in combustion, which can improve efficiency [28].

$$AFR = \frac{\dot{m}_{air}}{\dot{m}_{fuel}} \quad (3.1)$$

The amount of fuel injected per combustion stroke:

$$m_{fuel\ injection} = \frac{m_{air-fuel\ mixture}}{Scale\ Factor \times (AFR + 1)} \quad mg/stroke \quad (3.2)$$

In a complete 4-stroke cycle within the internal combustion engine (ICE), the crankshaft completes two revolutions, therefore the number of injections (combustion strokes):

$$N_{injection} = \frac{Engine\ RPM}{2 \times 60s} \quad times/s \quad (3.3)$$

The fuel mass flow rate in grams per second:

$$\dot{m}_{Fuel} = \frac{N_{injection} \times m_{fuel\ injection}}{1000} \quad g/s \quad (3.4)$$

For engine thermal efficiency, the chemical energy of the fuel injected per second is calculated based on the RON99 petrol supplied by the FS event, with petrol heat value of around 45 MJ/kg [29].

$$\eta_{thermal} = \frac{Engine\ power\ output}{Chemical\ energy\ of\ the\ fuel\ injected\ per\ second} \times 100\% \quad (3.5)$$

Due to the utilization of a simplified driver model in the LapSim Code, where the throttle level is limited to either 0% or 100%, fuel data from the ECU fuel map (Appendix Figure C.1) was extracted specifically at full throttle. These engine curves were subsequently incorporated into the LapSim code for both the KTM and Triumph Engines.

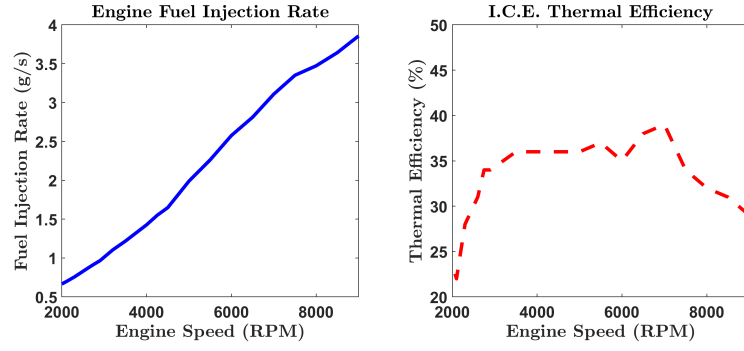


Figure 3: KTM690 Engine Fuel and thermal efficiency curve

3.2 Motor Modeling

In previous designs, BLDC motors have been employed, and it is observed that these motors generally experience a decrease in available torque and frequency as the load increases. Such a reduction in torque can adversely affect the motor's capacity to sustain speed and power output. However, the decision has been made to utilize a Permanent Magnet Synchronous Motor (PMSM) instead. PMSM motors offer improved speed control accuracy and exhibit lower torque ripple. This translates to a more consistent torque output across various speeds, resulting in smoother operation and reduced vibrations. As a consequence, modeling the power and torque curves of PMSM motors becomes more straightforward. The curves in Figure 4 have been extracted from the Emrax 188 motor datasheet [30], with blue line (60 kW) for peak running condition (less than 2 minutes) and red dash line (34 kW) for continuous running condition. The efficiency map [31] has also been extracted and given to the simulation team for motor thermal power modeling.

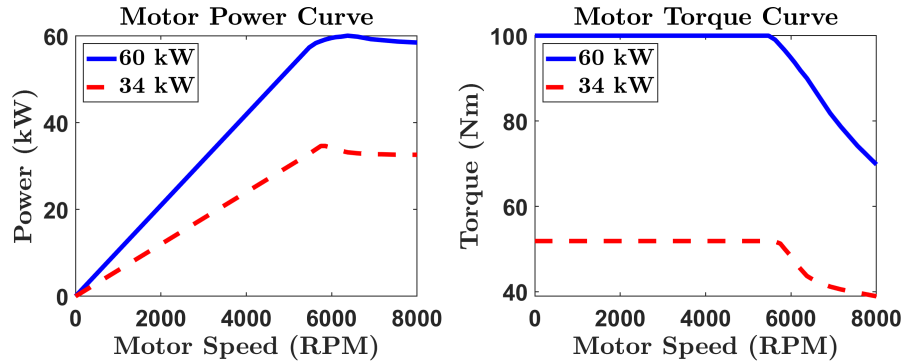


Figure 4: Emrax 188 HV LC motor power and torque curve

4 Engine and Motor Gear Ratio Tuning

As highlighted in previous GDP research, the tuning of the gear ratios involves a trade-off between the vehicle's top speed and torque at the wheels. Increasing the reduction ratio enhances torque but reduces the top speed.

The transmission system of the KTM 690 engine comprises a fixed primary transmission ratio connecting the engine output gear to the gearbox input gear, as well as a set of 6-speed gearbox ratios listed in the Table 5 below. It is important to note that none of these ratios can be altered without making modifications to the engine or gearbox. The ICE-Gearbox system is connected to the rear differential via a chain, and the transmission of the electric motor is designed as a two-stage system, with a planetary gearbox of reduction ratio 3.0 and a final drive chain, for more detail please refer to Shiven [13].

Table 5: KTM 690 Duke gear ratio [32]

Primary Ratio	6-speed MT Gearbox					
	G1	G2	G3	G4	G5	G6
79/36	35/14	28/16	28/21	23/21	22/23	20/23

The contribution to wheel torque originates from both the engine and motor, where τ denotes for torque and I denotes for transmission ratio:

$$\tau_{\text{axle-engine}} = \tau_{\text{engine}} \times I_{\text{primary ratio}} \times I_{\text{gearbox ratio G1-G6}} \times I_{\text{engine final ratio}} \quad (4.1a)$$

$$\tau_{\text{axle-motor}} = \tau_{\text{motor}} \times I_{\text{motor planetary gearbox}} \times I_{\text{motor final ratio}} \quad (4.1b)$$

Considering the P3 motor Hybrid layout chosen, the motor speed and wheel speed are directly linked at a fixed ratio. Additionally, the engine RPM ω_{motor} is associated with the wheel speed through six different ratios, which can be separated by disengaging the clutch to enable independent operation of the electric motor.

$$\omega_{\text{wheel}} = \omega_{\text{motor}} \times I_{\text{motor planetary gearbox}} \times I_{\text{motor final ratio}} \quad (4.2a)$$

$$= \omega_{\text{engine}} \times I_{\text{primary ratio}} \times I_{\text{gearbox ratio G1-G6}} \times I_{\text{engine final ratio}} \quad (4.2b)$$

As shown in Equation 4.3, the speed of the car is related to the tyre size, to maintain a same car speed, smaller tyre requires higher wheel speed. At the beginning of the gear ratio tinning, the design has been optimised for a 18 inch tyre. However, the aerodynamics team made the decision to reduce air resistance by reducing the tyre size D_{tyre} to 16.2 inches (0.41148 m) [2], resulting in new iterations being carried out in terms of final drive ratio tuning.

$$V_{\text{car}} = \frac{\pi D_{\text{tyre}} \cdot \omega_{\text{wheel}}}{60} \times 3.6 \quad \text{km/h} \quad (4.3)$$

As previously stated, the engine primary ratio, gearbox ratio, and the motor planetary gearbox ratio are not intended to be modified due to their complexity. This leaves the final ICE & Motor drive ratio tuning as the only adjustable parameter.

To evaluate the trade-off between top speed and high torque, a thorough analysis was conducted using the LapSim code. The results of this study are presented in the Table 6 below, providing insights into the impact of different final drive ratio settings.

Table 6: LapSim Results for different final drive ratios

Top Speed (km/h)	167	153	141	130	124	119	111	107	101
Engine Final Ratio	32/15	35/15	38/15	41/15	43/15	45/15	48/15	50/15	53/15
Motor Combined Ratio	3.7	4	4.4	4.8	5	5.2	5.6	5.8	6.1
Sprint time (s)	65.669	65.664	65.662	65.669	65.709	65.780	65.950	66.129	66.490
Acceleration time (s)	2.973	2.9701	2.971	2.993	3.021	3.055	3.121	3.167	3.240

Consequently, the range of optimal top speeds for achieving the fastest acceleration and shortest lap time in the sprint event falls between 141 km/h and 153 km/h. As a final design, the engine final ratio 48/19 (2.526) and combined motor ratio $3 \times 38/26 = 4.3846$ are selected due to sprocket constraints [13], which gives a top car speed about 141 km/h. Final Gear shifting map is shown in Figure 5.

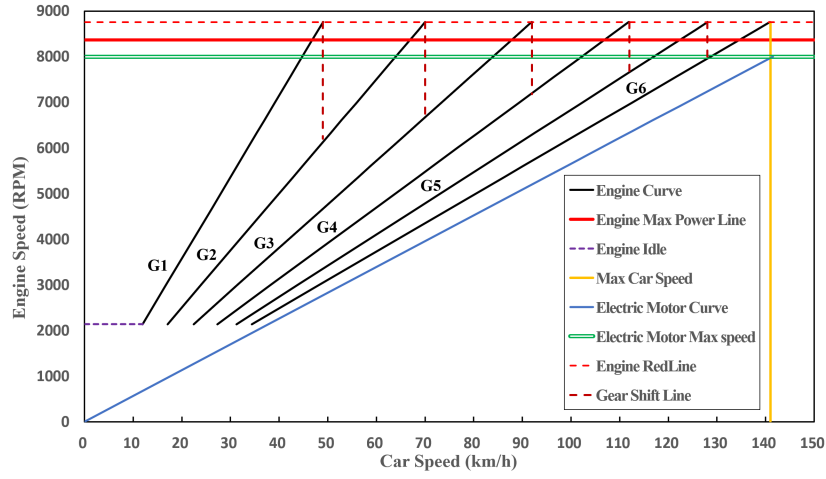


Figure 5: Final Gear Ratio Tuning and Shifting Map

5 Hybrid Strategy

5.1 Final Design

After conducting an investigation into various hybrid strategies outlined in Junjie Zhuge's report [9], a together strategy utilizing two motor power maps has been selected, considering their feasibility and ease of implementation, as well as their compatibility with LapSim modeling.

In the context of events where lap time is the sole scoring metric, a peak 60 kW output mode has been applied to the motor, allowing for the generation of maximum torque at the wheels.

Figure 6 shows the theoretical power and torque exerted on the rear axle, which is calculated based on the engine and motor curves, assuming no tire slip.

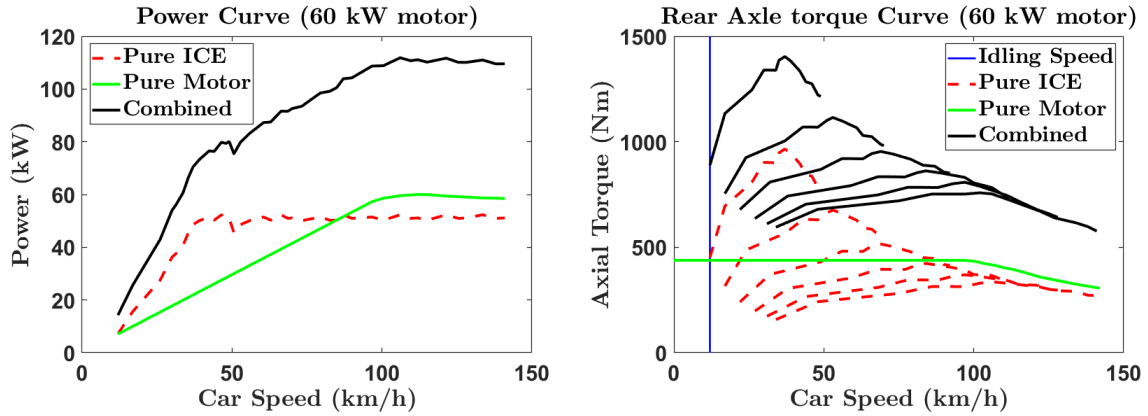


Figure 6: 60 kW Performance Mode Power and Torque Diagram

However, in light of energy consumption concerns during the endurance event, a lower 34 kW continuous motor output map with a cut-off at 4,800 rpm has been implemented, shown in Figure 7. This ECO mode combines the torque output of the motor with that of the ICE until a speed of 84.9 km/h, taking a balance between energy consumption and acceleration performance. For further details regarding the cut-off tuning, please refer to Aaditya's report from the Performance Modeling team [11].

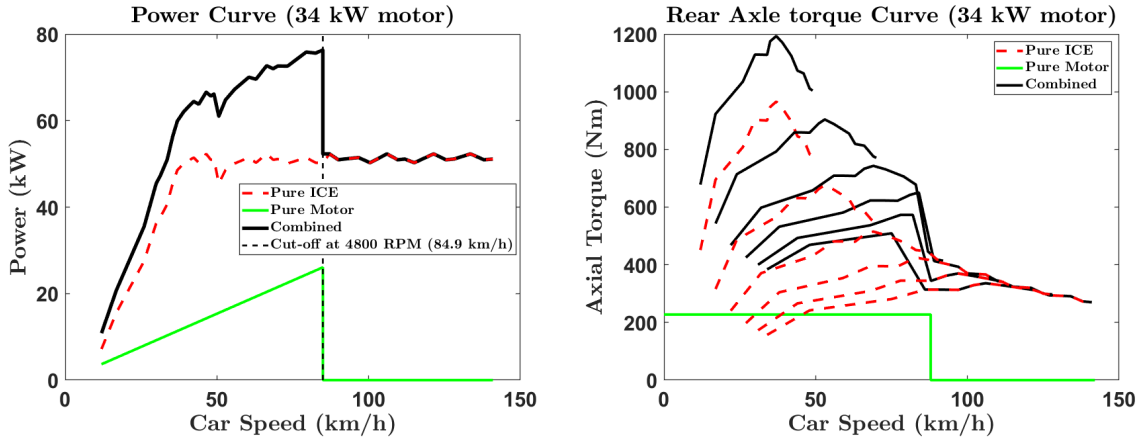


Figure 7: Power and Torque Diagram of 34 kW Eco Mode with 4,800 RPM motor cut-off

5.2 Hybrid System Improvements

During the endurance event, the implementation of the aforementioned Eco mode results in an electrical energy consumption of 4.14 kWh, corresponding to a theoretical battery cell weight of 15.94 kg [9] in the first iteration. Considering a 15% redundancy, the estimated battery weight may exceed 18 kg. This raises the possibility of further enhancing the hybrid system through two approaches:

1. Utilizing the engine to charge the battery during the engine's high-efficiency RPM range. By doing so, the required battery capacity can be reduced, resulting in weight reduction.
2. Implementing regenerative braking to capture and store the energy dissipated during braking. Since braking occurs frequently throughout the lap, this method not only reduces the battery mass& capacity requirement but also lowers the overall electrical energy consumption, thereby increasing the efficiency score.

Both of these methods can potentially be integrated into the HEV2023. By sending control signals to the Unitek BAMOCAR D3 motor controller, the Emrax 188 motor can be instructed to generate negative torque [33], leading to the generation of a backwards electromotive force (EMF) for charging the battery.

5.2.1 Engine Recharging

The calculation of CO_2 emissions in FSUK is derived from a value of 2.31 kg/L for petrol and 0.45 kg/kWh for electricity [14]. To evaluate the potential benefits of using the engine to charge the battery, an analysis was conducted. The gain from engine recharging was assessed for various engine RPM values:

$$\text{Gain} = CO_2 \text{ emission using engine recharge 1kwh electricity} - 0.45 \quad (5.1)$$

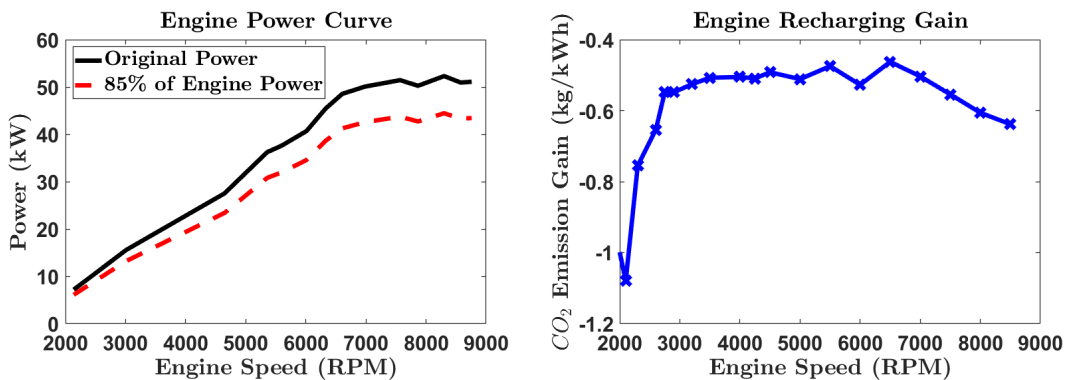


Figure 8: Analysis of the gain on engine recharging

However, as illustrated in Figure 8, it is evident that using the engine for recharging purposes results in a negative gain in CO_2 emissions. This indicates that it is more advantageous to store additional energy within the battery rather than relying on the engine to recharge it. Additionally, engine recharging draws power from the ICE, consequently reducing the overall power output of the vehicle. To exemplify this, a degraded power output curve is depicted in Figure 8, assuming a fixed extraction percentage of 15%. Based on these considerations, the decision has been made to abstain from utilizing engine recharging.

5.2.2 Regenerative Braking

Regenerative braking is a widely adopted technological approach to enhance efficiency in both commercial and racing vehicles. Once the gear ratio design has been finalized, the regenerative braking capability of the motor was investigated. The Emrax 188 motor theoretically has the potential to generate a maximum negative torque of 100 Nm [30] at 60 kW. However, this capacity may be limited by the battery's charging capability. In Junjie's report, a comprehensive convergence study was conducted to evaluate the relationship between battery charging capability and regenerative braking power, resulting in a suggested maximum regenerative braking power of 19.5 kW [9][11].

Considering the final combined motor ratio of 4.3846 [13], the regenerative torque capability of the motor is presented in Figure 9. Since the FSUK regulations only permit mechanical brake systems without electronic control [14], the regenerative torque is restricted based on the most limiting scenario, which is the top speed of the car. This secures a consistent motor brake behaviour at any speed and yields a maximum brake torque of 107.30 Nm on the rear axle and a maximum total regenerative brake force of 521.53 N on the rear wheels.

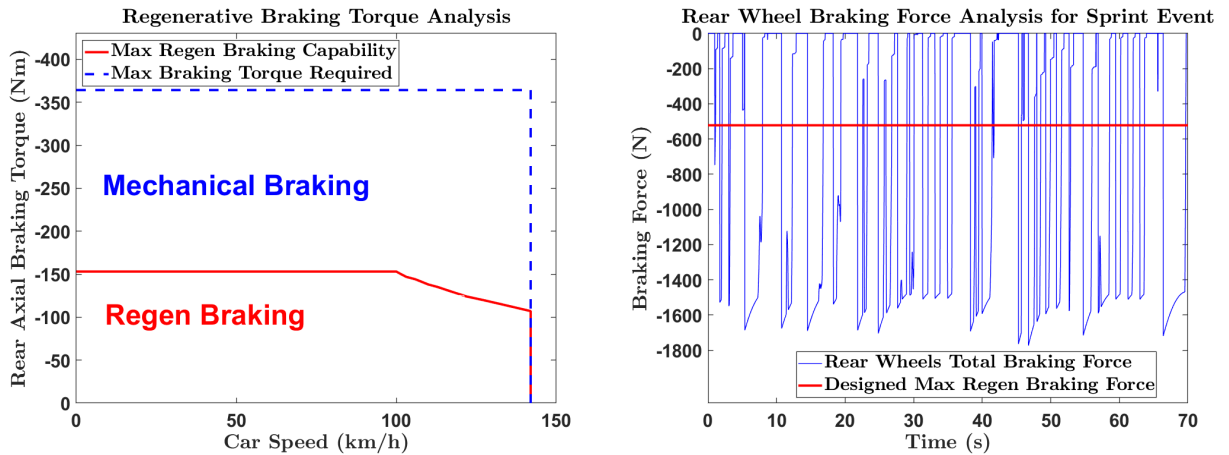


Figure 9: Regenerative braking capability analysis

Based on the LapSim results, each rear wheel requires a maximum brake force of 885.06 N, leaving an additional mechanical brake force of 625.80 N to be provided by the mechanical brake system, which was designed by Leihao Wang [34]. The final regenerative braking design ensures a linear brake behaviour suggested by the brake team, with around first 30% of the brake pedal travel to be regen braking only. In addition, an extra position sensor on the brake pedal is needed for detecting driver's brake input, which has been integrated by the electrical system Designer Jiatan Wang [35].

As a result, the implementation of regenerative braking has led to an improvement of 3 points in the efficiency score. Additionally, it has contributed to a reduction in the battery mass by approximately 3 kg [9]. Further details regarding the simulation results can be found in Junjie Zhuge's report.

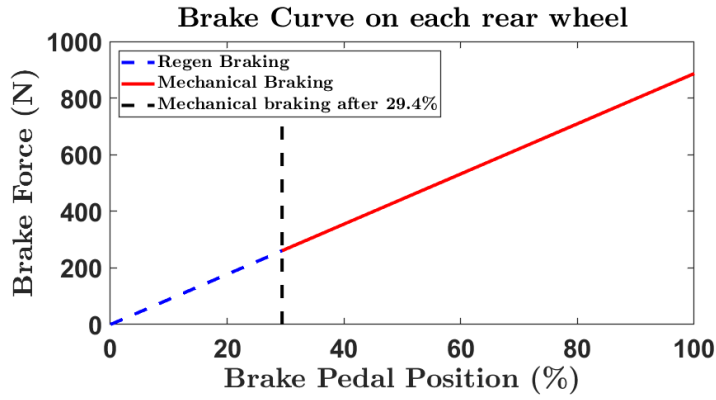


Figure 10: Brake Curve on each rear wheel

6 Traction Control

In the context of vehicle launch and driving at the corners during the lap, it is advantageous to maximize torque on the driving wheel while preventing wheel slippage to achieve the best possible acceleration and controllability. The data presented in Figure 6 indicates that the total combined torque can marginally exceed 1400 Nm. Based on LapSim analysis, the maximum traction force during acceleration, without any tire slippage, is measured at 3000.3 N on each rear wheel, resulting in a combined torque of 1234.6 Nm. This finding suggests the need to limit the torque output of the vehicle. To address this requirement, the previously chosen race-based Motec M800 ECU can be equipped with a Traction Control Multiplexer (TCMux) [36]. The electrical team has developed four-wheel speed sensors capable of relaying the driven and undriven speeds to the ECU. By specifying a target slip rate, the ECU can regulate the engine throttle to restrict the torque output accordingly. A similar approach can also be implemented in the motor controller.

7 Limitation and Further Improvements

The powertrain design was among the initial designs to be finalized in the HEV2023 project, allowing more time for the other teams to focus on their respective tasks. Unfortunately, due to delays in the progress of other team's developments, such as the aerodynamic kit, the final cut-off RPM was not finely tuned, thereby preventing the hybrid powertrain system from fully unleashing its potential to maximize the score.

Considering that the current engine curves are derived from the original KTM motorcycle, it implies that the power and throttle map are not optimized for our HEV2022 race car. Therefore, it is recommended to adjust the ECU power map specifically for this vehicle. However, there is a limitation within the LapSim's driver model, which assumes that the throttle input can only be either 0% or 100%. This restriction hampers the modeling of more advanced hybrid strategies. For instance, the motor usage and power output could be dependent on the required throttle level, rather than simply based on cut-off RPMs.

Presently, the structure only permits a parallel mode. However, if the engine generator were to be replaced by another P0 or P1 motor, it would enable a P0+P3 hybrid configuration, facilitating a mode switch between serial and parallel modes with greater potential. Additionally, in future investigations, if a clutch could be implemented on the motor side, it would potentially allow for a higher motor ratio, i.e. high motor torque, without the risk of motor over-spinning.

8 Conclusion

To summarize, this year's project has brought about a remarkable transformation compared to previous GDP designs. It features a fresh configuration and a complete redesign of components, resulting in a substantial reduction in the overall weight of the powertrain and a greater level of hybridization. This new configuration not only ensured that the HEV2023 achieved an impressive score of 522.7 points, securing it the first-place position in the event, but it also opens up exciting possibilities for future advancements and developments.

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Appendix

A 2022 HEV4 Curves

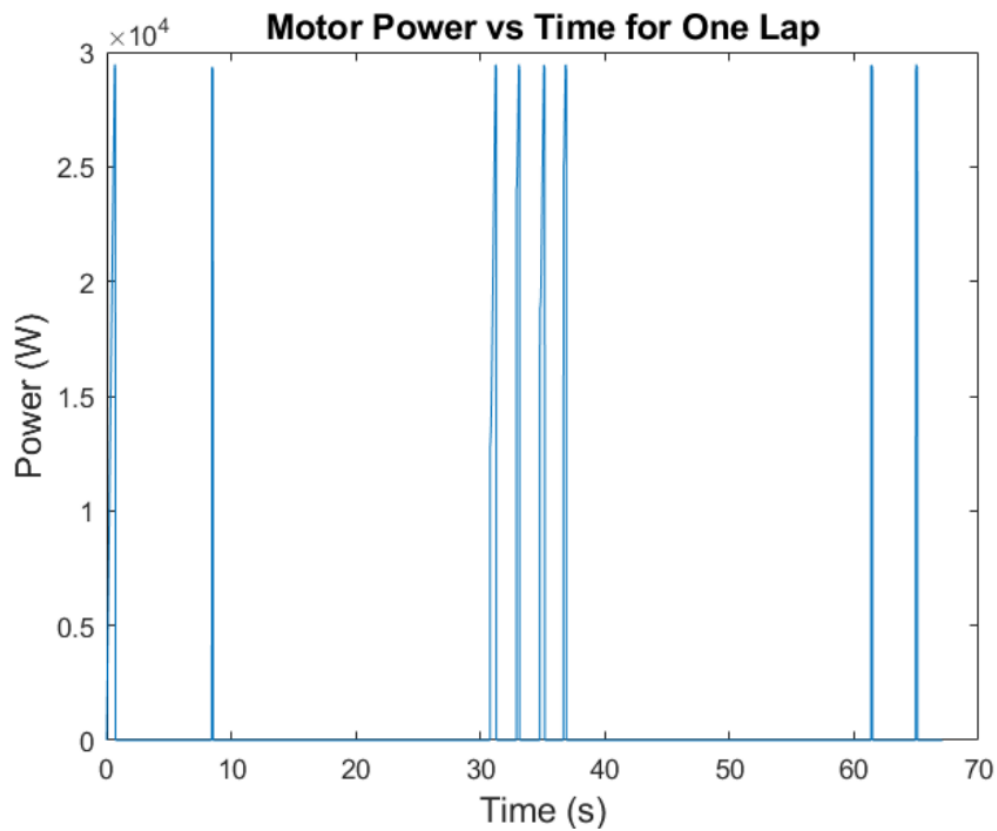
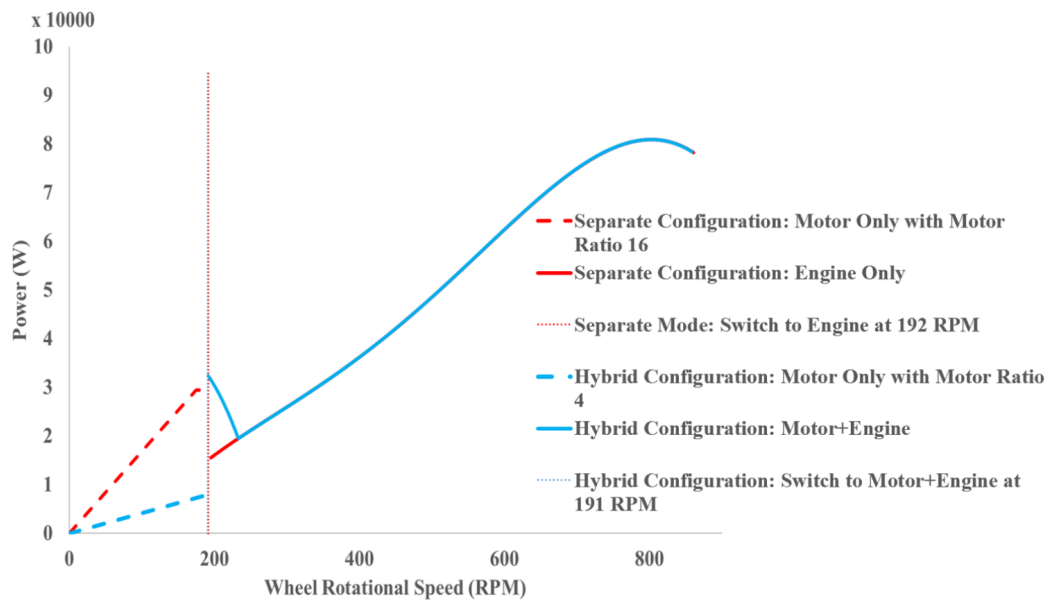


Figure A.1: 2022 GDP Hybrid Power Curve [4]

B Comparable teams in FSUK 2023

Comparable Vehicles

	 2022 FSUK Winner Glasgow UGR	 FS Team Uttejti, India	 MoRe Modena Racing Hybrid, Italy
Mass	Approx. 210 kg	Approx. 250 kg	250 kg (but with carbon fibre chassis)
Engine	Honda CBR600 L4 48 kW @ 8,700 rpm	KTM LC4 390cc 30 kW @ 9,600 rpm	Modified Ducati 478cc V2 46 kW @ 10,500 rpm
Motor	N/A	Approx. 30 kW	Customised 96v PMSM 32.5 kW (peak) @ 8600 rpm
Battery Pack	N/A	Unknown	Sony VTC6, 27s10p 97v 2.9kWh 30kW(max) Cell weight 13kg
Config	ICE RWD HF: 0	Hybrid RWD, Parallel HF: 0.5	Hybrid RWD, Parallel HF: 0.3-0.39
Power to Mass Ratio	Around 0.2	0.24	0.31 kW/kg
Results	493 points in FSUK 2022 1st overall	Formula Hybrid USA 2019 6th overall	Static events only in FSUK 2022

HF: Hybrid Factor = EM Power / (EM Power + ICE Power)

Figure B.1: Comparable teams analysis in FSUK 2023

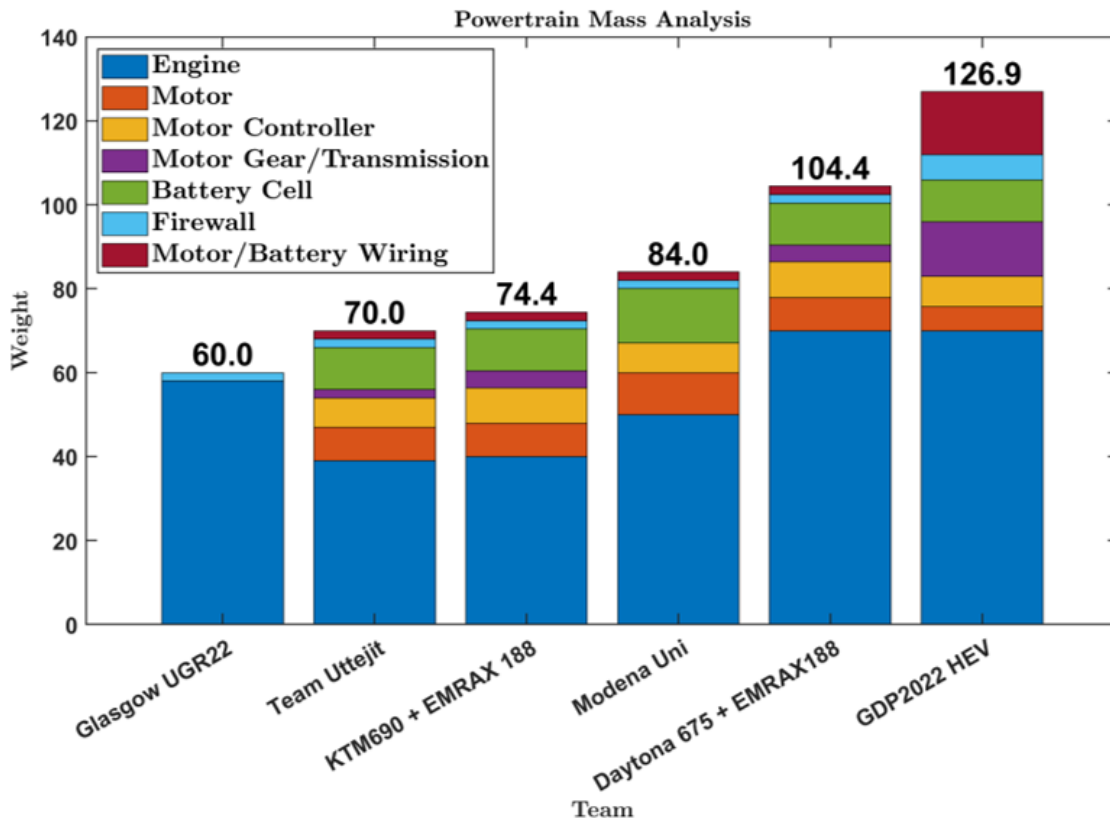


Figure B.2: Comparable teams powertrain mass analysis

C KTM 690 Engine data&curves

TuneECU (Unregistered)																	Map	Table	ECU	⋮
Rpm	Throttle Position (%)																			
	0	1	2	3	4	5	6	8	10	15	20	25	30	35	40	50	60	70	80	100
15000	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
11000	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
10500	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
10000	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
9500	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
9000	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
8500	1000	1000	1000	1000	1000	1022	1130	1457	1913	2391	2630	2870	3217	3826	4314	4978	5630	6525	8304	10800
8000	1000	1000	1000	1000	1000	1022	1189	1533	1932	2425	2745	2965	3283	3870	4435	5022	5652	6652	8393	10940
7500	1000	1000	1000	1000	1000	1022	1318	1659	2090	2515	2864	3206	3578	4221	4829	5467	6119	7221	8819	11260
7000	1000	1000	1000	1000	1001	1135	1577	1923	2332	2734	3033	3379	3739	4463	5177	5801	6553	7618	9235	11190
6500	1000	1000	1000	1000	1038	1205	1643	2017	2433	2850	3160	3442	3827	4699	5440	6152	6965	8106	9376	10900
6000	1000	1000	1000	1000	1033	1105	1349	1807	2195	2535	2962	3285	3594	3899	4740	5560	6295	7113	8220	9490
5500	1000	1000	1000	1000	1087	1171	1522	1999	2352	2687	3034	3358	3765	4141	4997	5838	6637	7493	8334	9475
5000	1000	1000	948	1094	1210	1622	2124	2449	2747	3138	3470	3913	4340	5201	6082	6908	7577	8236	9233	10030
4500	1000	1000	1036	1173	1418	1894	2316	2583	2864	3254	3610	4098	4564	5542	6485	7154	7805	8197	8713	9235
4250	1000	994	1080	1214	1512	1948	2403	2629	2887	3333	3688	4162	4628	5682	6646	7308	7745	8197	8663	9190
4000	1000	1000	1095	1295	1604	2041	2472	2670	2946	3362	3749	4209	4688	5783	6751	7338	7760	8136	8377	8994
3500	1000	1001	1251	1439	1823	2277	2614	2813	3080	3479	3909	4361	4860	6173	6984	7426	7715	7937	8247	8769
3200	1000	1116	1353	1590	1948	2391	2705	2918	3221	3752	4210	4774	5335	6443	7013	7428	7624	7903	8163	8696
2900	1000	1228	1526	1802	2139	2569	2807	3056	3419	3948	4506	5029	5742	6717	7109	7383	7513	7751	7953	8393
2750	1066	1358	1643	1943	2245	2627	2914	3146	3484	4137	4685	5327	5909	6771	7145	7383	7549	7739	7953	8405
2600	1202	1428	1749	2022	2355	2691	2997	3241	3603	4257	4811	5444	6040	6714	7143	7395	7502	7739	7941	8384
2300	1214	1523	1919	2181	2512	2878	3212	3421	3853	4595	5405	5986	6507	6821	7062	7373	7585	7799	8034	8301
2100	1368	1693	2041	2323	2638	2951	3283	3673	4202	5030	5829	6349	6761	7100	7318	7511	7729	7971	8141	8346
1900	1588	1940	2266	2538	2889	3174	3477	4010	4528	5124	5805	6325	6725	7064	7306	7499	7753	8008	8215	8431
1750	1668	2120	2435	2718	2997	3248	3534	3935	4412	5046	5619	6103	6452	6733	7026	7274	7533	7874	8146	8359
1650	1715	2210	2525	2795	3026	3260	3511	3785	4133	4737	5249	5707	6116	6402	6756	7070	7329	7615	7942	8269
1550	1783	2289	2593	2855	3020	3243	3482	3705	3970	4424	4977	5333	5802	6197	6552	6906	7233	7451	7819	8106
1200	2150	2652	2792	2915	3020	3193	3358	3540	3722	3986	4486	4763	4994	5407	5843	6266	6651	6944	7383	7792
1000	2310	2695	2798	2918	3020	3177	3342	3490	3656	3870	4276	4647	4895	5225	5506	5853	6261	6568	6966	7526
800	2310	2689	2809	2929	3026	3193	3358	3474	3623	3804	4255	4564	4812	5110	5423	5787	6171	6532	6930	7510
0	2200	2430	2596	2713	2896	3160	3342	3474	3606	3788	3986	4449	4713	5010	5308	5688	6081	6442	6875	7477

Figure C.1: KTM 690 ECU data [27]

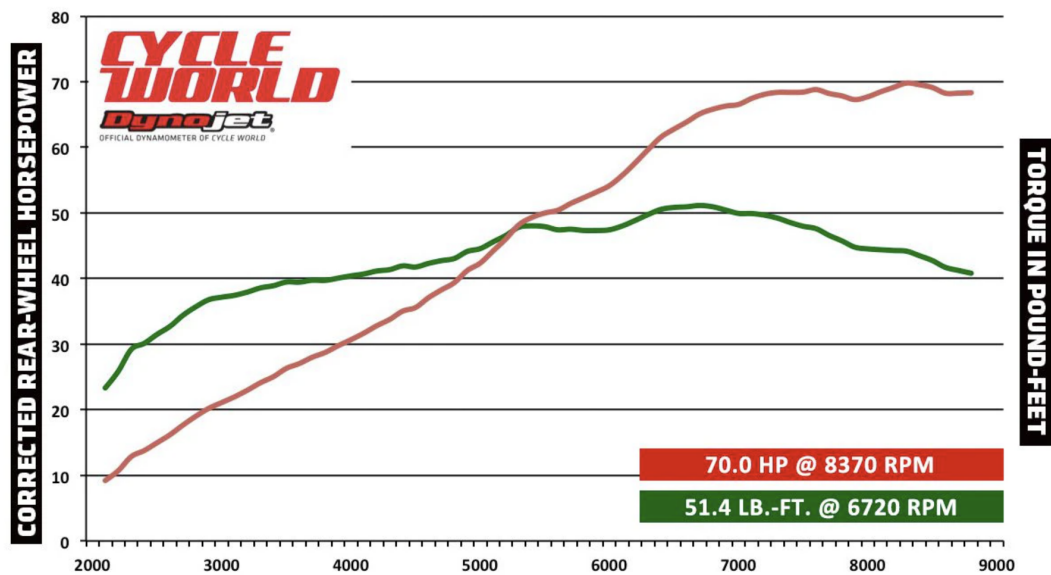


Figure C.2: KTM 690 Duke Dyno Curve

D Hybrid Powertrain Final Designed Power Curves

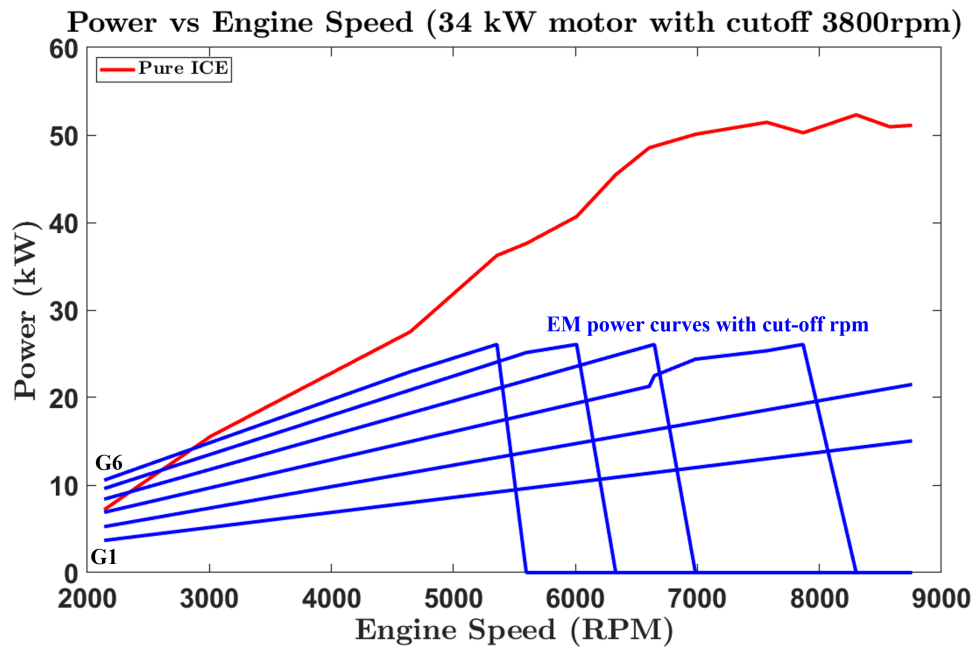


Figure D.1: Eco Mode ICE and EM Power curve vs engine RPM

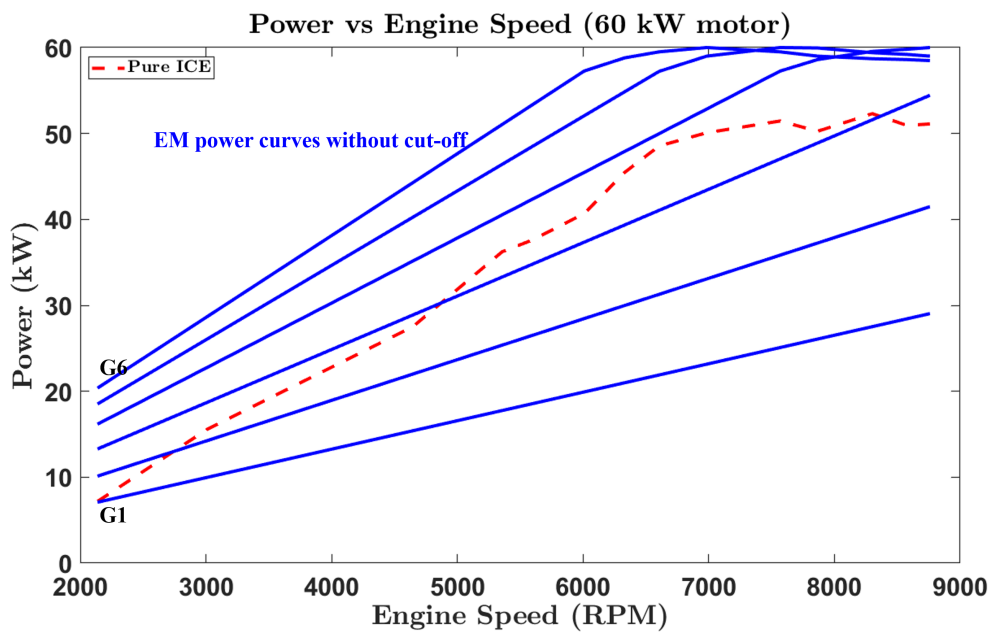


Figure D.2: Performance Mode ICE and EM Power curve vs engine RPM